

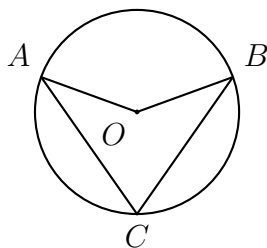
Multi-Theorem Circle Configurations

Inscribed angles, arcs and sectors, segment relationships, and circle equations

Directions

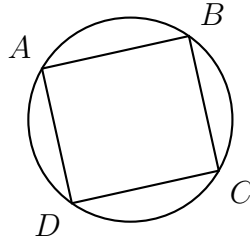
- Most problems need *two* theorems. Name each theorem you use beside the line where you use it.
- The diagrams show the configuration only — every measurement you need is in the words of the problem, and the diagrams are not drawn to scale.
- Leave π in your working and round only at the last step, to the precision the problem asks for.

1. In circle O , points A and B lie on the circle and the central angle $\angle AOB$ measures 110° . Point C lies on the major arc AB . Find $m\angle ACB$.



Answer: _____ degrees

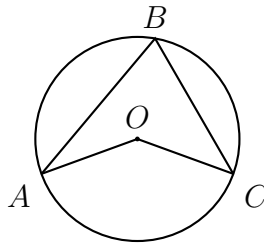
2. Quadrilateral $ABCD$ is inscribed in a circle, with $m\angle A = 84^\circ$ and $m\angle B = 71^\circ$. Find $m\angle C$ and then $m\angle D$.



Answer: _____ degrees

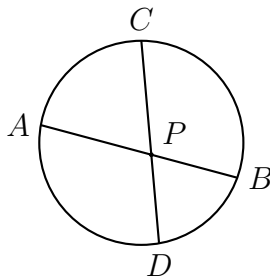
Answer: _____ degrees

3. In circle O the radius is 9 cm and the inscribed angle $\angle ABC$ measures 40° . Find the length of the minor arc AC , to the nearest tenth of a centimetre.



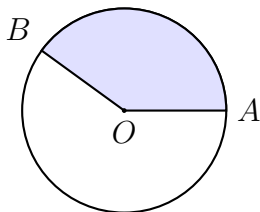
Answer: _____ cm

4. Chords AB and CD of a circle meet at the interior point P . The four pieces measure $AP = 8$, $PB = x$, $CP = 6$ and $PD = 12$. Write the equation the intersecting-chords theorem gives, then solve it for x .



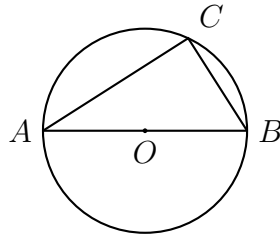
Answer: _____

5. A circular flowerbed of radius 10 cm is drawn on a plan. Sector AOB has central angle $\angle AOB = 144^\circ$. Find the area of sector AOB , to the nearest tenth of a square centimetre.



Answer: _____ cm^2

- 6.** In circle O , chord AB is a diameter and C is another point on the circle. In triangle ABC , $m\angle ABC = 34^\circ$. Find $m\angle BAC$.

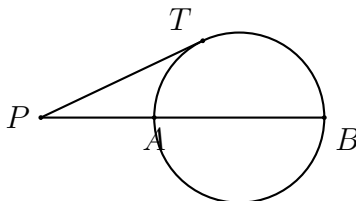


Answer: _____ degrees

- 7.** A circle has equation $x^2 + y^2 - 10x - 6y + 9 = 0$.
- (a) Complete the square to write the equation in centre-radius form, and state the centre and the radius.
- (b) Find the two x -values where this circle crosses the x -axis.

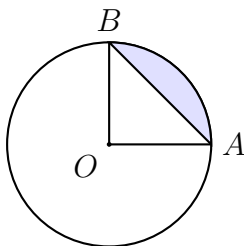
Answer: _____

8. From an exterior point P , the tangent segment PT touches a circle at T , and a secant from P passes through the circle meeting it first at A and then at B . If $PT = 12$ and $PA = 8$, find the length $AB = x$.



Answer: _____

9. In circle O the radius is 14 cm and the central angle $\angle AOB$ is a right angle. Find the area of the shaded circular segment cut off by chord AB (the sector minus the triangle), to the nearest tenth of a square centimetre.



Answer: _____ cm^2

10. A circular fountain is modelled on a site plan by $x^2 + y^2 - 8x + 6y - 11 = 0$, with both coordinates in metres.

- (a) Find the centre of the fountain and its radius.
- (b) A sprinkler at the centre waters a sector with central angle 120° . Find the area it waters, to the nearest tenth of a square metre.
- (c) Explain why the watered area is that fraction of the whole fountain's area.

Answer: _____ m

Answer: _____ m^2